

Raghumani Mehta

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Education

University of British Columbia

Vancouver, BC

B.Sc. Computer Science (Major), Minor in Physics

Expected Graduation: 2027

Academic Standing: 86.9% Cumulative Average | **Dean's List** (2024–2025)

Coursework: Computer Networking (A+), Algorithms (A+), Databases (A+), Software Eng (A), Computer Systems (A), Operating Systems (A-), Computer Vision (A-).

Technical Skills

- **Languages:** Python, C++, C, Rust, Java, TypeScript, JavaScript, SQL
- **Frameworks & Tools:** Node.js, Express, Spring Boot, Git, Docker, ROS, GitHub Actions
- **Cloud & DevOps:** AWS (Textract, Lambda, S3, SQS, SNS, RDS), Linux/Unix, POSIX
- **Testing:** JUnit, Mocha/Chai, Pytest, GDB, Valgrind

Technical Work Experience

UBC Sailbot | github.com/UBCSailbot/sailbot_workspace

Feb 2025 – Present

Pathfinding Software Team Lead (Python, ROS, Docker)

Lead since Sept 2025

- Lead 4 developers in weekly scrums; own pathfinding architecture, code reviews, and CI pipelines.
- Optimize route-planning algorithms under uncertain conditions to improve simulated sail-time efficiency.
- Build mock testing frameworks to validate autonomous navigation logic prior to hardware deployment.
- Coordinate autonomy design decisions with electrical and mechanical engineering leads.

UBC Department of Computer Science

Sept 2024 – April 2026

Undergraduate Teaching Assistant

- **CPSC 317: Computer Networking** (Jan 2026 – April 2026)
Incoming TA; selected to support students with socket programming and transport layer protocols.
- **CPSC 310: Software Engineering** (Sept 2024 – Dec 2025)
Mentored 50+ students on Node.js/TypeScript architecture; resolved complex Yarn and CI/CD issues.

Technical Projects

Reliable UDP | github.com/raghumanimehta/ReliableUDP

Jun 2025 – Present

Systems Project (C++, Linux/Unix Sockets)

- Engineering a reliable data transfer layer over UDP to replicate TCP guarantees (ACKs, sequence no.s).
- Developing custom sender/receiver components utilizing POSIX networking APIs for robust delivery.
- Handling edge cases including packet loss and out-of-order delivery to ensure 100% data integrity.

mycp Utility | github.com/raghumanimehta/mycp

May 2025 – Jun 2025

Systems Project (C, POSIX System Calls)

- Recreated Linux cp utility using low-level POSIX system calls (`read`, `write`) and buffered I/O.
- Managed file descriptors and permission flags to understand and replicate OS-level error handling.

OCR System | github.com/CPSC319-2025/Team-3-CCS

Jan 2025 – Apr 2025

Backend Project (Python, AWS, PostgreSQL)

- Designed a serverless backend pipeline using AWS Textract, Lambda, and SQS for parallel processing.
- Architected a high-availability system supporting 50+ concurrent users with zero data loss.

Leadership

Jump Start Orientation Leader | UBC CSDL

Aug 2024 – Sept 2024

- Mentored 25 first-year students and led a time-management seminar for 100+ attendees.